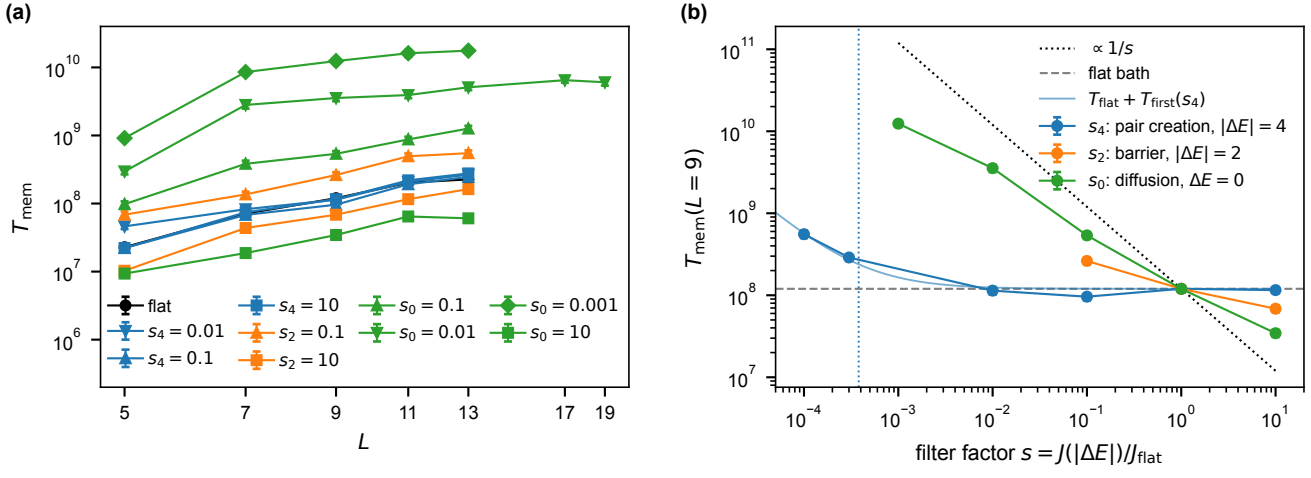
(a)
T_{mem}
10^{10}
10^{9}
10^{8}
10^{7}
10^{6}
5
7
9
11
13
17
19
L
flat
$s_4 = 0.01$
$s_4 = 0.1$
$s_4 = 10$
$s_2 = 0.1$
$s_2 = 10$
$s_0 = 0.1$
$s_0 = 0.01$
$s_0 = 0.001$
$s_0 = 10$
(b)
$T_{\mathrm{mem}}(L = 9)$
10^{11}
10^{10}
10^{9}
10^{8}
10^{7}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
filter factor $s = J(|\Delta E|)/J_{\mathrm{flat}}$
$\propto 1/s$
flat bath
$T_{\mathrm{flat}} + T_{\mathrm{first}}(s_4)$
s_4: pair creation, $|\Delta E| = 4$
s_2: barrier, $|\Delta E| = 2$
s_0: diffusion, $\Delta E = 0$